

POPOVICI GABRIEL-SORIN

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PROFESSIONAL OVERVIEW: Versatile **Full-stack Senior Software Engineer** and **Certified Software Architect (CPSA-F)** with over 15 years of experience in the **Java** ecosystem, C/C++, **JavaScript**, **SQL**, **Cloud-native** architectures (Kubernetes/Openshift). Proven track record delivering large-scale distributed systems in Internet Domains & Hosting, Civil Aviation, Finance/Banking, Oil & Gas and Public Sector industries.

Skills

Programming Languages: Java | Kotlin | JavaScript | C++ | SQL | Python

Frameworks: Spring Boot | Spring Framework | Reactor | Java EE (CDI/Weld) | React | Angular | Apache Camel | Camunda | Spring Integration | Spring Security | Spring Data

Databases: PostgreSQL | MariaDB | Oracle | Redis | MongoDB | Couchbase | Elasticsearch

Messaging: Kafka | IBM MQ | ActiveMQ

Cloud & DevOps: Kubernetes | OpenShift | Docker | Istio | Jenkins | TeamCity | Prometheus | Grafana | ELK | MinIO

APIs & Protocols: REST | OpenAPI/Swagger | FIX | FpML | IBML | Protobuf

Architecture: Distributed Systems | Event-Driven Design | Design Patterns | Enterprise Integration Patterns | Reactive Systems

Agile & Quality: Scrum | TDD | BDD | CI/CD | JUnit | SonarQube

AI & Developer Assistants: Copilot

Languages

English: C1

Italian: B2

German: A2

EXPERIENCE

Application Architect/Senior Engineer

July 2023 - Present

Employer: IONOS Internet Development Romania, 246C Calea Floreasca, 16th Floor, District 1, Bucuresti 014476, Romania

Mail & Media projects and tasks:

- **Android Mobile Development/Mail Cloud & News Application (January 2026-present):** worked on the cloud-storage application enabling users to browse, manage files, uploads/download content, organize photos in a timeline, automatically backup selected folders, create/share links, preview media, track background transfers. Led some large Java to Kotlin transformations and Spec Driven Development implementations using OpenSpec AI agentic tooling. *Tech stack:* Android/Kotlin 2.4.x, Java 17, UI: Google Jetpack Compose, Material 3, internal Nova design, Dagger 2, Retrofit, OkHttp, RxJava, Room, Gradle Kotlin DSL, SonarQube, Sentry, Junit, Mockito, Roboelectric
- **Enterprise Architecture & Modernization:** led the initiative of a top-down target architecture redesign for Mail Domain Hosting modernization using Microservices, Spring Boot 3,

and Kubernetes and written comprehensive technical documentation including ARC42 (ADRs|TCs)

- **AI & LLM Innovation:** Developed AI-powered business intelligence tools using Python, LangChain, and LLMs (Qwen/Llama/GPT) to analyze iOS/Play Store feedback. Engineered a microsite generator leveraging Ollama's Qwen Vision model to automate website creation for small businesses.
- **Frontend Modernization:** developed the "Light Control Panel," a mobile-friendly application built with React 19, TypeScript, Tailwind CSS, Vite, improving user accessibility.
- **Distributed Systems Design:** Designed and implemented critical microservices including the Mail Service (email aliases/distribution lists), Onboarding Service (conversion workflows), SSL Service (certificate management), and NIS2 Event Handler for regulatory compliance.
- **Security & Infrastructure:** Implemented zero-trust security using mTLS via Istio Service Mesh and OAuth2. Orchestrated event-driven architectures using Apache Kafka and db-scheduler.

Technologies: Java 21+, Spring Boot 3.x, Spring Security, Spring Data, Spring AI, Python, LangChain, Apache Wicket, React 19, TypeScript, Kubernetes, Istio, Kafka, Redis, Docker, MariaDB, Spring Modulith, Liquibase, Quartz, db-scheduler

Role: Senior Engineer

March 2019 – July 2023

IONOS projects and tasks:

- **Cloud Migration Strategy:** implemented the migration of legacy Domains platform applications to OpenShift cloud infrastructure. Implemented CI/CD pipelines using Jenkins and Helm, and established monitoring stacks with Prometheus, Grafana, and ELK.
- **API Design & Development:** Designed and implemented OAS-compliant public APIs for WebHosting Product Services (WHPSS) and Domain Customer Interaction Systems, enabling end-users to manage web spaces (WordPress, Drupal), DNS security, and domain transfers.
- **Big Data Integration:** Engineered the "Domains Expiry Pool" system to process large-scale ICANN TLD files using Apache Spark and Redis, integrating data with AI recommendation engines from InternetX.
- **Designed and implemented a public-facing OpenAPI (OAS) Specification** for SMB customers and tenants, enabling programmatic management of their entire domain portfolio (DNS, security, transfers, contacts). Engineered robust authentication and authorization flows by integrating directly with the Keystone Identity Service.
- **Cloud-Native API Architecture:** Designed and implemented comprehensive OpenAPI (OAS) specifications for the Domain Customer and Interaction System APIs on OpenShift. Engineered robust endpoints for single and bulk domain operations (lifecycle management, DNS security, transfers, renewals) and integrated with Keystone Identity Service for secure authentication/authorization and the Provisioning Service System (PSS) for backend fulfillment.
- **Workflow Modernization:** Executed a Proof of Concept (PoC) to migrate legacy J2EE BPMN engines to cloud-native solutions using Camunda/Activiti, contributing deployment descriptors to the open-source community(<https://github.com/popovici-gabriel/activiti7-openshift/tree/master/setup>).
- **High-Performance Backend:** Developed reactive microservices using Spring WebFlux and Reactor, optimizing for non-blocking I/O operations with MongoDB and PostgreSQL.

Technologies: Java 11, Spring Boot, Spring WebFlux, Spring Cloud Config, Spring Boot Admin, Apache Spark, OpenShift, Jenkins, Elasticsearch (ELK), Prometheus, PostgreSQL, MongoDB,

Couchbase, Swagger/OAS, S3 Min.io, OpenCSV, Maven/Gradle, Kafka, Camunda 7x. BPMN , Activity 5.x, Signavio , Quartz

Senior Engineer

May 2018 - March 2019

Employer: Luxoft Romania, Murex connectivity project

Project and tasks:

- **Financial Data Integration:** Architected and implemented a critical content distribution adapter bridging the Bloomberg Market Data real-time platform with Murex trading services, enabling seamless import of high-frequency pricing data.
- **Compliance & Security:** Integrated tightly controlled consumption (DDM Local-Publish) and entitlement management (EMRS) leveraging the Bloomberg Open API (BLAPI) for secure, compliant data access.
- **Cloud Connectivity:** Engineered "Zero Footprint" support for cloud deployments, ensuring robust functionality for remote and hybrid environments.

Technologies: Java 8, Java EE CDI API 1.x, Jboss Weld 2.x, Arquillian Weld, Camel 2.x , Google Guava, Bloomberg Open APIs BLAPI, Maven, Murex APIs, Cucumber, Fitness

Senior Associate Developer

Sep 2015 - April 2018

Employer: JPMorgan Chase & Co UK, Investor Services, CIB, Prime Brokerage, 35 Waterloo St, Glasgow Technology Center G2 6HS, www.jpmorganchase.com

Projects and tasks:

- **Microservices Migration (CFS):** Architected and executed the decomposition of a legacy Equity Swaps monolith into manageable Microservices, enabling scalable Contract for Difference (CFD) processing and trade lifecycle management using FpML/IBML logical data modeling.
- **Automated Client Reporting:** Designed and built a configurable, on-demand reporting feeder utilizing Spring Integration and Enterprise Integration Patterns, supporting multiple protocols (SFTP, AMPS) for diverse downstream recipients.
- **High-Frequency Trading (GRIPS):** Support the "Global Real Time Position Keeping Service" (GRIPS) for low-latency equity swap execution. Leveraged Pivotal Gemfire to create an in-memory allocation pool, automating replenishment triggers based on real-time utilization thresholds.
- **System Optimization & Caching:** Implemented distributed in-memory caching using Hazelcast and Gemfire to optimize Equity Swaps queries and enhance the corporate action workflow engine.
- **Data Integration & Quality:** Integrated market data (Reference Data Technology) via REST/OAS and NoSQL, and connected the Equity Swaps platform with middle-office accounting systems using FIX protocol and Angular 2. Introduced SonarQube and automated regression testing (BDD) to ensure code quality and continuous delivery.

Technologies: Java 6/7/8, Java EE (legacy platforms using JBoss), Spring 4.x framework (Core, Boot, JavaConfig, JDBC, Integration, Data, Retry, Cache, JMS, MVC), Spring Statemachine, Spring Pivotal Cloud Foundry, Spring Cloud Config, Angular 2.x with TypeScript, Mule ESB, Pivotal Gemfire, Hazelcast, Oracle Database, JSON, XML/FpML/IBML, Google Protocol Buffers v2.0, IBM MQ, HornetQ, ActiveMQ, AMPS (from www.crankuptheamps.com), Cucumber, Jacoco, Cobertura, Apache PIT, Google Guava, Oracle Database 12c, Jersey, Swagger 2.0 OAS, Springfox, Junit 4.x/5.x, Jackson, Vavr, CI/CD pipelines using TeamCity, Git/Bitbucket, IntelliJ IDEA Ultimate Edition, Microsoft Visual Studio Code

Senior Software Engineer

Mar 2014 - Aug 2015

Employer: *COMPUTAS IT* Romanian subsidiary of foreign-owned systems integration and software development outsourcing company (2012 net sales in the range of millions of EUR, payroll less than 100), Bucharest, www.computas.ro

Projects and tasks:

- Logistics hub: automated tracking of goods within the supply chain of onshore/offshore facilities of oil companies
- Architected and developed UI web application using XHTML, CSS, jQuery, XSL Transformations and Xpath; implemented REST-full services based on EPCIS - EPC Information Services Standard and ISO 15926 industrial automation standards for: Cargo Containing Unit's, Organization, Hire, Journey, Leg, Observations, Documents, Events model; developed Templates, UI Components using SPARQL Web Pages for Legs, Journeys and Observations data models.

Technologies: javascript, jQuery, XSLT, XSD, XML, XHTML, Java 1.7, Apache Tomcat 7.x, Apache Maven 3.x, Apache CXF 2.7.x/ JAX-RS 2.0 - JSR 339, Google Guava, Apache Commons; Semantic: SPARQL, RDF/RDFS, SWP developed with TopBraid Composer Maestro Edition-IDE, TopBraid Live Enterprise Server, Oracle 12c with RDF support

Java Software Engineer

Aug 2013 – Feb 2014

Employer: KEBA Automation SRL, Romanian subsidiary of foreign-owned industrial, banking & energy automation company-Romanian subsidiary develops proprietary software for foreign company-manufacturer of automation equipment (2012 net sales lower than 10 Mil. EUR, payroll of less than 50 employees), Bucharest, www.keba.com

Projects and tasks: developed middleware software applications to be used with banking automation equipment. Performed detailed analysis of the incidents reported by customers, transactions reconciliation and application errors. Created bug reports based on reported incidents and applications log files. Developed proprietary communication protocol for banking terminals. Developed customized ATM transactions software for foreign and Romanian banks. Integrated 3rd party payment/money transfer systems-Western Union. Improved Maven build scripts for automatic deployment into several production environments.

Technologies used: Java 1.6, C# .Net 2.5, ActiveX, JavaScript, Bash shell scripting, Maven, Spring Core/MVC/ORM/JDBC/AOP/Security/JMS, Jasper Reports, Hibernate 3.x, Swing, Netbeans IDE, Netbean Visual Library API, JMS ActiveMQ, IBM MQ 6.x, Oracle, MySql, H2, Derby, Apache Commons, Google Guava, JUnit, dbUnit, EasyMock, Mockito, XStream, XML-Apis, JAXB, JAX-WS, SoapUI.

Freelance Java Software Engineer

Apr 2013 – Aug 2013

Contractors: *Anritsu Srl*, Italian subsidiary of foreign-owned measuring instrument manufacturer, Rome-Italy (project-based related to the engineering of measuring instruments proprietary software, 4-month project); Romanian systems integrator-one of Top 5 companies in segment (project-based work on a Client Relationship Management-CRM application for a major power utility company, 4-month project), Bucharest, www.anritsu.com

Tasks: performed reverse engineering work of the Hadoop version 1.x IPC low level protocol to C/C++, and integrated Hadoop into middleware MasterClaw engine; related to the CRM application: developed custom made modules of standardized files, designed portal user interface, integrated application with 3rd party SAP/Energy products. Published on Git-Hub for the open source community under <https://github.com/popovici-gabriel/projects.git>

Technologies: Java 1.7, JAXB, Oracle Server, SAP Java Connector 2.x, Hibernate 3.x, Spring MVC, Spring ORM, Spring JPA, Spring JDBC, Apache Wicket, Apache POI, Apache iText, Jasper Reports, Google Guava

Java Software Engineer

Dec 2011 - Feb 2013

Employer: Sabre Holdings, Airline Solutions US-based proprietary software development company-travel industry, USA-based job under H1B visa status

Projects and tasks: performed active software engineering role in the development of a complex online booking engine/e-commerce application.

Technologies: Java SE 1.6, Spring Core/WebFlow/ORM/JPA/JDBC, Apache CXF, junit, Apache Commons, Hibernate.

Software Engineer/Systems Administrator

Jan 2011 - Nov 2011

Employer: Z-Terra Inc. US-based proprietary software development company for the oil exploration and production industry, US-based job under H1B visa status, www.z-terra.com

Tasks: provided technology support for employees and customers; managed small computer farm (50+ Unix/Linux systems) of high performance computer data centers; developed Z-Eos Seismic data inspection: GUI application viewer(2D) for seismic data.

Technologies: C/C++, Qt SDK, Qwt Project, Glib, lib PVM, cmake/qmake, Qt Creator, IDE, Eclipse CDT, SQLite, Stanford Exploration Lib - SEPLib.

Software Engineer

Nov 2007 - Dec 2011

Employer: Canadian Bank Note International Europe, Romanian subsidiary of one of the largest specialized foreign-owned printing companies worldwide, Bucharest (Romanian subsidiary develops specialized software applications for governments worldwide), www.cbnco.com

Projects: vehicle registration portal, driver license issuing system, passport issuing control systems, border management systems.

Tasks: implemented dynamic report generation using Apache POI, engineered change requests, improved error handling and logging mechanism, developed web application scheduler, developed centralized task scheduler mechanism using JAX-WS SOAP web services and java 1.5 concurrency API, engineering change requests, dynamic PDF generation, UI front end development.

Technologies: Microsoft Visual Studio 2005, Eclipse, Java 6, Hibernate, Crystal Reports, Apache XML-RPC, Servlet 2.4, Tomcat 6.x, Rhino, OpenSCDP, MFC, Apache Axis 1.4, Servlets 2.4, JSP 2.0, JSTL, Struts 1.3.10, Java Mail, Hibernate, Crystal Reports, Tomcat 5.x, Log4j, Ant, Java 5 SE, JSP/Servlets, JSTL, Javascript, iText, JAX-WS, Struts 1.2.9.

Junior Software Developer

Aug 2006 - May 2007

Employer: Canadian Bank Note International Europe, Romanian subsidiary of one of the largest specialized foreign-owned printing companies worldwide, Bucharest (Romanian subsidiary develops specialized software applications for governments worldwide)

Projects: vehicle registration portal and vehicle technical inspection application

Tasks: web application development, bug fixing, unit testing

Technologies: Java SE 1.5, Java Servlets/JSP, Struts 1.2.9, Hibernate, Javascript, junit, Tomcat 5.x, Ant

EDUCATION

Diploma Engineer, University „POLITEHNICA” of Bucharest, Faculty of Automatic Control and Computers, 2001-2006, Bucharest, Romania.

HONORS

“Leonardo da Vinci” Mobility Programme RO/2006/PL97046/S “Modern techniques for automatic identification”, Department of Computer Science, University of Turin, Italy, April-June 2007

IT CERTIFICATIONS

- Certified Professional for Software Architecture Foundation Level, 2025
- Sun Certified Business Component Developer Java EE Platform 5, CX 310-091, 2010
- Sun Certified Developer for Java Web Services, CX – 310-220, 2008
- Sun Certified Web Component Developer for the Java Platform, Enterprise Edition 5, 2008
- Sun Certified Programmer for the Java Platform, Standard Edition 6.0, CX-310-065, 2008

IT TRAININGS

- Architecting on AWS, Global Knowledge 4502, November 2017, Glasgow UK
- Developing on AWS, Global Knowledge 4504, November 2017, Glasgow UK
- Creating Distibuted Applications Using Pig and Hive on Hadoop, ROITraining 435, November 2017, Glasgow UK